

Candidate Name: _____

Class	Index No



Preliminary Examination 2008
Pre-University 3

MATHEMATICS
Higher 2

9740/02

Wednesday

10 Sep

3 hours

Additional materials:
Cover Page
Writing paper

READ THESE INSTRUCTIONS FIRST

Write your name, index number and class in the spaces at the top of this page and on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagrams or graphs.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use a graphic calculator.

Unsupported answers from a graphic calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphic calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

The number of marks is given in brackets [] at the end of each question or part question. At the end of the examination, fasten all your work securely together.

This question paper consists of **6** printed pages.

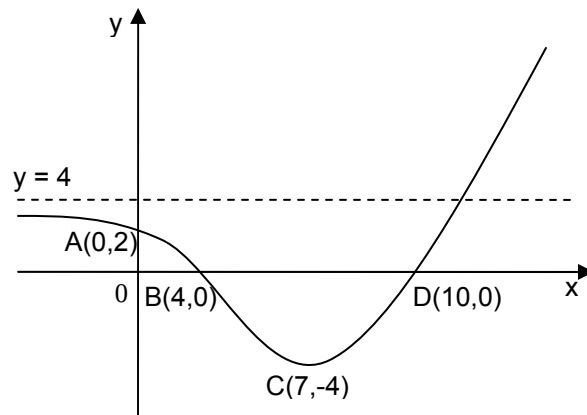
[Turn over]

Section A: Pure Mathematics [40 marks]

1 (i) Solve the inequality $\frac{9x+2}{x-2} \geq 2$, leaving your answer in exact form. [3]

(ii) Hence, solve the inequality $\frac{9e^x+2}{2-e^x} \leq -2$. [2]

- 2 The following diagram shows the graph of $y = f(x)$. The curve passes through the points A(0,2), B(4,0), C(7, -4), and D(10,0), and has an asymptote $y = 4$.



On separate diagrams, sketch the graphs of

(i) $y = |f(2x)|$, [2]

(ii) $y = \frac{1}{f(x)}$, [3]

indicating clearly the points corresponding to A, B, C, D where appropriate.

- 3 (i) By repeated differentiation of $\sin 2x + \cos 2x$, find Maclaurin's series for $\sin 2x + \cos 2x$, up to and including the term in x^3 . [4]

- (ii) Hence, obtain the expansion of $\sin^2 2x + \frac{1}{2} \sin 4x$ up to and including the term in x^3 . [3]

- (iii) State the range of values of x for which the expansion is valid. [1]

- 4 (a) The sum of the first n terms of a series is given by $S_n = n^2 - 2n$. Find the n^{th} term in terms of n , and hence show that the series is an arithmetic progression. [3]

- (b) Watermelons are harvested from a farm on a monthly basis. As the fertility of the land decreases over time, the quantity of watermelons harvested decreases with each harvest. Each subsequent month's harvest contains 1% less watermelons than the previous month. In the first month, the crop contains 2,000 watermelons. Calculate the maximum number of watermelons which can be harvested from this farm in the long run. [2]

The owner of the farm decides to stop his harvesting operations when the crop yields less than 500 watermelons a month. The month where he harvests less than 500 watermelons will be his last month of harvesting. Find the total number of months of harvesting. Given that the first harvest was in January 2008, when will the last harvest be? [5]

- 5 The functions f and g are defined by

$$f : x \mapsto \frac{3x}{x+2}, \quad x > -2,$$
$$g : x \mapsto x^2 + 1, \quad x \leq 5.$$

- (i) Find the equations of the asymptotes of $y = f(x)$. [2]
- (ii) Sketch the graph of $y = f(x)$. [3]
- (iii) Determine, stating your reasons, whether each of the following functions exists:

(a) g^{-1} ,

(b) gf .

If any of the functions exists, find its rule, domain and range. [7]

[Turn over]

Section B: Statistics [60 marks]

- 6 Give a real-life example of a scenario where systematic sampling could be used, and briefly describe how it can be carried out. [3]
State one reason why this could be preferred over stratified sampling, in relation to the scenario you have described. [1]

- 7 (a) A particular social gathering consists of 40 people. Each person makes a handshake with every other person in the gathering. Find the total number of handshakes made. [2]

- (b) Another gathering consists of 5 pairs of married couples. The entire group stands in a line, with each man standing next to his wife. Find the number of different possible arrangements in which this can be done. [3]

- 8 A newspaper claims that newly-graduated Economics students have an average starting salary of \$2600. The director of a recruitment company believes that the figure stated in the newspaper is incorrect, and does a survey with a group of 80 newly-graduated Economics students.

The data the director obtained is as follows, with \$x representing the monthly salaries of these graduates

$$\sum (x - 2500) = 6400, \quad \sum (x - 2500)^2 = 780,000.$$

Calculate the unbiased estimates of the mean and variance of the monthly salary of a newly-graduated Economics student. [3]

Carry out an appropriate test at the 1% significance level to determine whether the director's belief is justified. [4]

- 9 A wildlife scientist found that the number of tigers in a particular jungle can be modelled by a Poisson distribution with mean 0.3 tigers per square kilometre. Find the probability that there are no tigers in a given jungle section of area 6 square kilometres. [2]

Using a suitable approximation, find the probability that, in a jungle section of area 100 square kilometres, there will be more than 25 but at most 40 tigers. [6]

- 10** Research is being carried out into how the population of a particular species of bacteria varies with temperature. Experiments give rise to the following data:

Temperature (x °C)	-4.1	-3.2	-1.8	-0.7	0.1	1.1	2.0	3.1	3.8
Population (y , in millions)	11.5	15.0	18.3	20.4	20.1	19.4	17.9	15.0	12.0

- (i) Give a sketch of the scatter diagram for the given data. [1]
- (ii) Without any calculation, comment on the correlation between temperature and the population of the bacteria. [1]
- (iii) State, with a reason, which of the following models is appropriate for the data collected.
- A:** $y = ax + b$,
- B:** $y = ax^2 + b$,
- C:** $y = a + \frac{b}{x}$,
- D:** $y = a + b \ln x$. [2]
- (iv) Based on the model you have chosen in (iii), calculate the values of a and b using the least squares method. [2]
- (v) Calculate an estimate of the population of bacteria when the temperature is 0.5°C . Comment on the reliability of your estimate. [3]
- 11 (a)** Events A and B are such that $P(A) = \frac{1}{6}$, $P(B' | A) = \frac{2}{3}$ and $P(A' \cap B') = \frac{2}{9}$.
- (i) Show that $P(A \cap B) = \frac{1}{18}$. [2]
- (ii) Find $P(B)$. [1]
- (iii) Are A and B independent events? Justify your answer. [2]

[Turn over]

- (b) Alicia, Britney, and Catherine decided to meet for lunch, but all three of them forgot if the venue was the Tuckshop, the Café, or the Fast Food Restaurant. Alicia throws a fair coin to decide between the Tuckshop and the Café. Britney does the same to decide between the Café and the Fast Food Restaurant. Catherine does the same too, but if she gets a head, she will go to the Tuckshop, while if she gets a tail, she will throw the coin again to decide between the Café and the Fast Food Restaurant.

Find the probability that

- (i) Britney and Catherine meet, [2]
- (ii) all three girls meet, [1]
- (iii) all three girls head to different places, [2]
- (iv) all three girls head to different places, given that Britney goes to the Café. [2]

- 12 The heights, in metres, of female and male students in a particular school follow normal distributions with the following data

	Mean Height (metres)	Variance (metres ²)
Female Students	1.6	0.04
Male Students	μ	σ^2

It is known that 20% of the males are shorter than 1.6m, while 75% of them are below 1.8m. Show that μ and σ are 1.71m and 0.132m respectively. [3]

Find the probability that

- (i) the heights of two randomly selected female students both exceed 1.55m, [2]
- (ii) the total height of two randomly selected female students is more than 3m, [3]
- (iii) a randomly selected male student is taller than a randomly selected female student by at least 20cm. [3]

A random sample of 20 female students is chosen. Find the probability that at least 15 of these female students are shorter than 1.7m. [4]

End of Paper